

Proof. **1. Trivial (base) cases.**

- **Case $n = 0$.**

$$(1 + x)^0 = 1 = 1 + 0 \cdot x.$$

- **Case $x = 0$.**

$$(1 + 0)^n = 1 = 1 + n \cdot 0.$$

- **Case $x = -1$.** Here $1 + x = 0$.

– If $n = 0$, then $(1 + x)^0 = 1 = 1 - 0$.

– If $n = 1$, then $0 = 1 - 1$.

– If $n \geq 2$, then $(1 + x)^n = 0$ while $1 + nx = 1 - n \leq 0$, so $0 \geq 1 - n$.

2. Induction hypothesis. Assume that for a fixed integer $n \geq 1$,

$$(1 + x)^n \geq 1 + nx \quad \text{for all } x \geq -1. \quad (\text{IH})$$

3. Inductive step. Let $x \geq -1$, whence $1 + x \geq 0$. Multiplying (??) by the non-negative factor $1 + x$ preserves the inequality:

$$(1 + x)^n(1 + x) \geq (1 + nx)(1 + x). \quad (1)$$

The left-hand side is $(1 + x)^{n+1}$. Expanding the right-hand side gives

$$(1 + nx)(1 + x) = 1 + (n + 1)x + nx^2. \quad (2)$$

Substituting (??) into (??) we obtain

$$(1 + x)^{n+1} \geq 1 + (n + 1)x + nx^2. \quad (3)$$

Since $n \geq 1$ and $x^2 \geq 0$, the term nx^2 is non-negative; dropping it can only decrease the right-hand side:

$$(1 + x)^{n+1} \geq 1 + (n + 1)x. \quad (4)$$

Equation (??) is Bernoulli's inequality for the exponent $n + 1$.

4. Completion of the induction. The inequality holds for the base case $n = 0$, and the implication

$$(1 + x)^n \geq 1 + nx \implies (1 + x)^{n+1} \geq 1 + (n + 1)x$$

has been proved. By the principle of mathematical induction,

$$(1 + x)^n \geq 1 + nx \quad \text{for every integer } n \geq 0 \text{ and every } x \geq -1.$$

5. Equality cases. Equality may occur only at the steps where we did not lose a non-negative term.

- In the base case $n = 0$ we have $(1 + x)^0 = 1 = 1 + 0x$ for all $x \geq -1$.
- In the base case $x = 0$ we have $(1 + 0)^n = 1 = 1 + n \cdot 0$ for all $n \geq 0$.
- In the inductive step the discarded term is nx^2 . If $n \geq 1$ and $x \neq 0$, then $nx^2 > 0$, so the inequality in (??) is strict and hence (??) is strict. If $x = 0$, then $nx^2 = 0$ and equality is preserved for every n . If $x = -1$, then $1 + x = 0$ and $(1 + x)^n = 0$, $1 + nx = 1 - n$. Equality holds exactly when $n = 1$ (both sides equal 0); for $n \geq 2$ we have $0 > 1 - n$.

Collecting these observations, equality holds precisely in the following situations:

Condition	Reason
$n = 0$ (any $x \geq -1$)	Both sides equal 1.
$x = 0$ (any $n \geq 0$)	Both sides equal 1.
$(x, n) = (-1, 1)$	Both sides equal 0.

No other pair (x, n) with $x \geq -1$, $n \geq 0$ yields equality.

6. Conclusion. For every real $x \geq -1$ and every integer $n \geq 0$,

$$(1 + x)^n \geq 1 + nx,$$

with equality exactly in the three cases listed above. $\square$

$\square$